

WiFi Deauth Attack

■ Outcomes

- 2.1 Monitor and analyze WiFi traffic
- 2.2 Create a managed WiFi network
- 2.3 Perform and analyze a deauth attack on a WiFi network

■ Background

In this lab you will run the aircrack-ng tools on a Mininet-WiFi VM image demonstrate a WiFi deauth attack.

■ Installing aircrack-ng

Start the Mininet-WiFi VM the same way you did in the [Getting Started with Mininet-WiFi](#) and open a terminal.

Run `sudo apt install aircrack-ng` and hit enter or 'y' when prompted to install the packages.

■ Creating Traffic

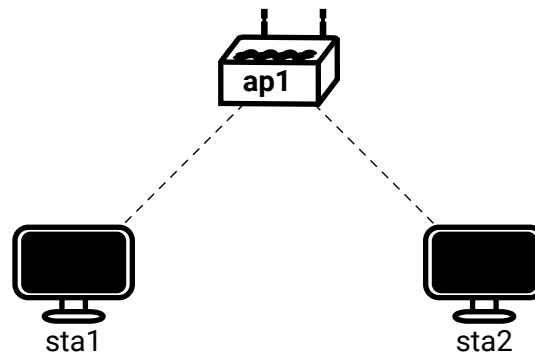


Figure 1: Mininet-WiFi Default Topology

Start the default Mininet-WiFi topology by running `sudo mn --wifi` (the wifi user password is wifi)

Now find the IPv4 address for `sta2` by running `sta2 ip address show dev sta2-wlan0`. Don't forget this address! Remember for an IPv4 address you're looking for 4 dotted decimal integers between 0 and 255. Don't include the subnet mask and don't use the broadcast address.

Now run `xterm sta1`. This will open a new terminal window as if it were running on `sta1`. Now inside the new xterm window run `ping <ADDRESS>` where is the IPv4 address of `sta2` that you found previously.

You now have `sta1` continuously pinging `sta2` through `ap1` in a simulated WiFi network.

■ Sending Deauth Frames

From Mininet-WiFi CLI (the one with the mininet-wifi prompt, not the xterm on `sta1`) run the following command to determine what the MAC address of the WiFi interface on `ap1` is: `ap1 ip link show ap1-wlan1`. Remember that MAC addresses are typically six hexadecimal bytes separated by colons (':'). Remember this MAC address, it is the BSSID for this WiFi network.

Again from the Mininet-WiFi CLI run the following command to determine what the MAC address of the WiFi interface on `sta1` is: `sta1 ip link show sta1-wlan0`. We will call this MAC address STA1-MAC.

Now run `xterm sta2` to open a terminal window as if it were running on *sta2*. *sta2* will be our attacker, we will send a deauth frame to disconnect *sta1* from *ap1*. Run the following command in this new terminal:

```
aireplay-ng --deauth=1 -a <BSSID> -c <STA1-MAC>
```

■ Analysis

Now check the other window. What responses are your pings getting? You can stop the ping with Ctrl-C and run `iw dev sta1-wlan0 link` to see the status of your connection to *ap1*.

■ Questions

1. What is the connection status of *sta1* and *sta2* at the end of this lab?
2. [Why might an attacker use a deauth attack on a WiFi network?](#)
3. Which part of the [CIA triad](#) is being targeted by this attack?
4. What mitigation techniques are there for this attack?

When you are done you can close the xterms, type exit in the Mininet-WiFi command line, and Shutdown the machine from the main menu.